

Multiplexed 7-Segment Display Using 74F157, 7448, and 74HC139

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Contents

1	Introduction	3
2	System Overview	3
3	Truth Table of BCD Inputs	3
4	74F157 Multiplexer	4
4.1	74F157 Image Placeholder	4
4.2	Function	4
4.3	Important Pins	4
4.4	Input Connections	4
4.5	Output Connections	5
4.6	Operation	5
5	7448 BCD to 7-Segment Decoder	6
5.1	7448 Image Placeholder	6
5.2	7448 Truth Table	6
5.3	Function	7
5.4	Input Pins	7
5.5	Segment Outputs	7
5.6	Operation	7
6	74HC139 Decoder	8
6.1	74HC139 Image Placeholder	8
6.2	Function	8
6.3	Important Pins	8
6.4	Connections	9
6.5	Operation	9
7	Complete System Operation	10
8	Advantages of Multiplexing	10
9	Conclusion	10

1 Introduction

This project demonstrates a multiplexed two-digit 7-segment display system using:

- 74F157 Quad 2-to-1 Multiplexer
- 7448 BCD to 7-Segment Decoder
- 74HC139 Decoder/Demultiplexer
- Two Common Cathode 7-Segment Displays

The circuit allows displaying two different BCD numbers on two 7-segment displays using only one BCD decoder.

2 System Overview

The system operation is divided into three main stages:

1. The 74F157 Multiplexer selects one of two BCD inputs.
2. The 7448 converts the selected BCD value into 7-segment signals.
3. The 74HC139 enables one display at a time for multiplexing.

3 Truth Table of BCD Inputs

The following table shows the BCD input combinations and the displayed decimal number.

Decimal	D	C	B	A	Binary
0	0	0	0	0	0000
1	0	0	0	1	0001
2	0	0	1	0	0010
3	0	0	1	1	0011
4	0	1	0	0	0100
5	0	1	0	1	0101
6	0	1	1	0	0110
7	0	1	1	1	0111
8	1	0	0	0	1000
9	1	0	0	1	1001

4 74F157 Multiplexer

4.1 74F157 Image Placeholder

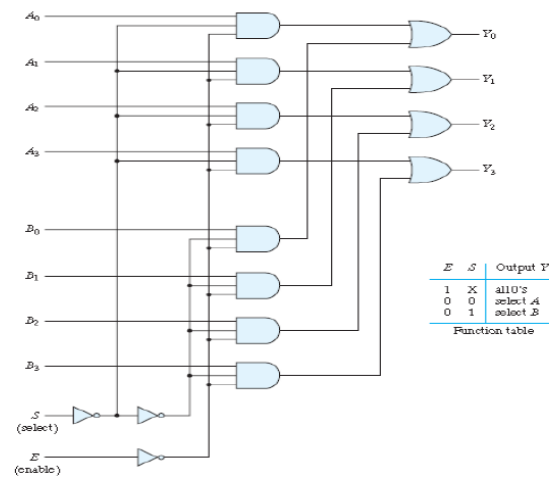


Figure 1: 74F157 Multiplexer

4.2 Function

The 74F157 is a Quad 2-to-1 Multiplexer. It selects one of two 4-bit BCD inputs according to the Select signal.

4.3 Important Pins

Pin	Function
1	Select (S)
15	Enable ($\bar{G}$)
16	Vcc (+5V)
8	GND

4.4 Input Connections

74F157 Pin	Connection
2	A0
3	B0
5	A1
6	B1
11	A2
10	B2
14	A3
13	B3

4.5 Output Connections

74F157 Output	Connected to 7448
Pin 4	A
Pin 7	B
Pin 9	C
Pin 12	D

4.6 Operation

- If $\text{Select} = 0$, the A inputs are selected.
- If $\text{Select} = 1$, the B inputs are selected.

Example:

- A Inputs = 0011 (Decimal 3)
- B Inputs = 0111 (Decimal 7)

Then:

- $\text{Select} = 0 \Rightarrow \text{Display}3$
- $\text{Select} = 1 \Rightarrow \text{Display}7$

5 7448 BCD to 7-Segment Decoder

5.1 7448 Image Placeholder

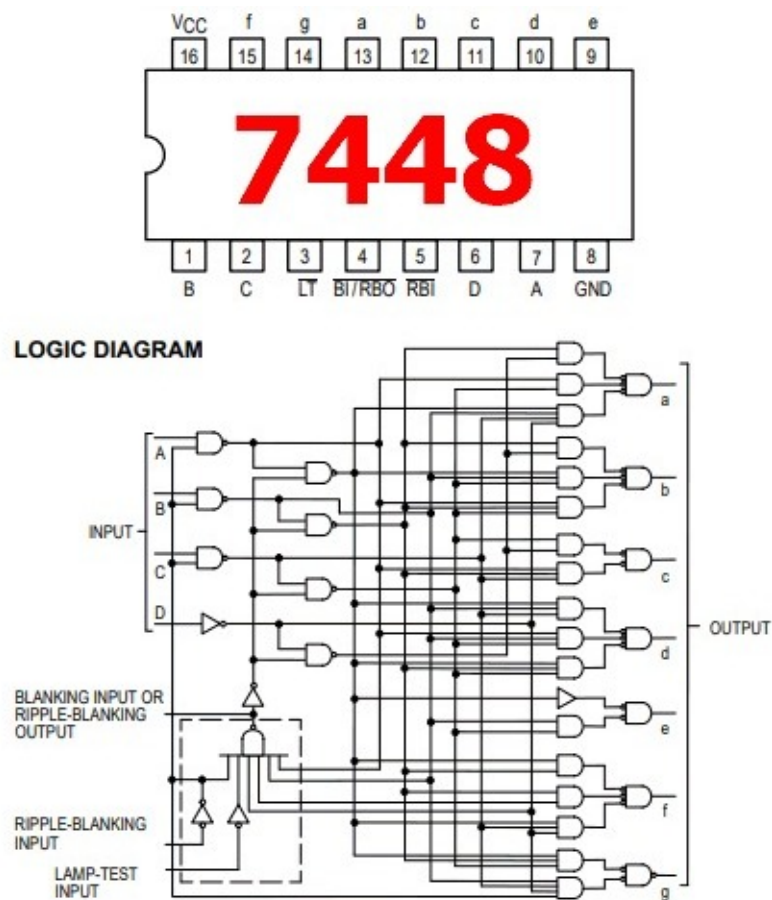


Figure 2: 7448 BCD to 7-Segment Decoder

5.2 7448 Truth Table

Decimal	D	C	B	A	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

5.3 Function

The 7448 converts a 4-bit BCD number into signals required to drive a Common Cathode 7-segment display.

5.4 Input Pins

7448 Pin	Function
7	A
1	B
2	C
6	D

5.5 Segment Outputs

7448 Pin	Segment
13	a
12	b
11	c
10	d
9	e
15	f
14	g

Each segment output is connected to the corresponding segment of both 7-segment displays through current-limiting resistors.

5.6 Operation

The decoder automatically lights the required segments according to the BCD input.

Example:

BCD Input:

0101

Displayed Number:

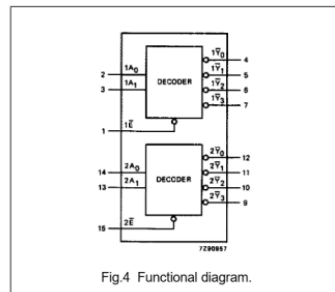
5

6 74HC139 Decoder

6.1 74HC139 Image Placeholder

Dual 2-to-4 line decoder/demultiplexer

74HC/HCT139



FUNCTION TABLE

INPUTS			OUTPUTS			
$\overline{nE}$	nA_0	nA_1	$\overline{nY_0}$	$\overline{nY_1}$	$\overline{nY_2}$	$\overline{nY_3}$
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	H	L	H	L	H	H
L	L	H	H	H	L	H
L	H	H	H	H	H	L

Notes

1. H = HIGH voltage level
L = LOW voltage level
X = don't care

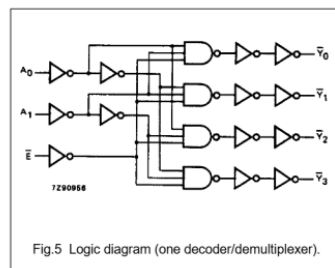


Figure 3: 74HC139 Decoder

6.2 Function

The 74HC139 is used for multiplexing between the two 7-segment displays.
It enables only one display at a time.

6.3 Important Pins

Pin	Function
1	Enable (G)
2	A Input
3	B Input
4	Y0 Output
5	Y1 Output
16	Vcc
8	GND

6.4 Connections

- Y0 is connected to the Common Cathode of the first display.
- Y1 is connected to the Common Cathode of the second display.
- Input A receives the same Select signal used by the 74F157.

6.5 Operation

- $\text{Select} = 0 \Rightarrow \textit{Firstdisplayenabled}.$
- $\text{Select} = 1 \Rightarrow \textit{Seconddisplayenabled}.$

The switching occurs very quickly, creating the illusion that both displays are active simultaneously.

7 Complete System Operation

1. The switches generate two BCD numbers.
2. The 74F157 selects one BCD number.
3. The selected number is sent to the 7448.
4. The 7448 generates the required segment outputs.
5. The 74HC139 enables the corresponding display.
6. Rapid switching creates multiplexed operation.

8 Advantages of Multiplexing

- Reduces hardware usage.
- Uses only one BCD decoder for two displays.
- Reduces power consumption.
- Simplifies the circuit.

9 Conclusion

This project demonstrates an efficient method for driving multiple 7-segment displays using multiplexing techniques.

The combination of the 74F157, 7448, and 74HC139 provides a simple and effective hardware design suitable for embedded systems and digital electronics applications.